Mid Sweden University (Mittuniversitetet)

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

DT168G / IU127G
Human-Computer Interaction / Interaction Design

Project group: Group 1

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Degree programme(s): [TO CONFIRM]
Examiner: [TO CONFIRM]

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Abstract

TODO: Write an abstract of approximately 250 words after the results and discussion are complete. It should summarise the project's background, purpose, method, results and conclusions.

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Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. Its concept sheet identifies students who regularly use music, podcast or audiobook applications as the intended users, including listeners who use several content types and those who mainly listen to music [1, p. 1]. The concept treats losing track of saved or unfinished content as a working problem to investigate through the design and evaluation of the interface.

The high-fidelity prototype is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two documented task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook [2, Sec. "Testable user flows"]. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The completed project should provide an identified set of user needs and a prioritised set of interface requirements. The requirements should make clear which listening activities the interface is intended to support and how they relate to the information collected from the target group. This provides a basis for assessing the design beyond its visual appearance.

The design deliverables should include documented alternatives, low-fidelity sketches and interactive high-fidelity prototypes. Alternatives should cover the organisation of audio categories and the presentation of content, such as category tabs compared with a mixed view with filters, and cards compared with lists. The sketches should preserve the early ideas, while the V1 and V2 prototypes should make the selected task flows available for examination. Together, these materials should show how the interface developed and why particular solutions were chosen.

A further deliverable is a documented usability evaluation of the important tasks: finding and saving content, locating saved content and returning to unfinished spoken audio. The evaluation should provide observations of task completion and difficulties, together with an account of how those observations informed design decisions. A record of design iterations and reasoned conclusions should connect the requirements, prototypes and evaluation. These goals concern identifiable project outputs; they do not assume that all proposed changes will prove effective.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or

subscription processing, licensing systems, backend databases or production account management. This scope is consistent with the prototype documentation, which describes a front-end-only implementation [2, opening description]. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

V2 provides a further prototype for examining changes to the interface. Its documentation explicitly describes these changes as redesign hypotheses whose effectiveness has not yet been evaluated [4, Sec. “Iteration 2 redesign”]. The existence of a revised prototype consequently does not, by itself, establish an improvement in usability.

1.5 Division of Work

The project group consists of Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad. The Project Session 2 data collection plan assigns Yacoub Dawli the interviewer role and assigns Obid Alrhman Aljabi and Sarshar Chavoshi Rad the recording of handwritten interview notes [5, p. 2]. This documents the planned allocation of those roles. It does not confirm which interviews were completed or who carried out the later design, implementation and evaluation work.

TODO – group confirmation: Record each member’s actual contributions to data collection and analysis, sketches, V1 and V2, usability-test planning and execution, and report writing. Confirm whether the planned interview roles were followed, and identify the report sections written or revised by each member before submission.

Chapter 2

Theory

TODO: Chapter not yet written.

Chapter 3

Method

TODO: Chapter not yet written.

Chapter 4

Design

TODO: Chapter not yet written.

Chapter 5

Results

TODO: Chapter not yet written.

Chapter 6

Discussion

TODO: Chapter not yet written.

References

[1] O. A. Aljabi, Y. Dawli, and S. Chavoshi Rad, “Omnisound: Project session 4 concept sheet,” Unpublished project document, Group 1, completed final version, single page.

[2] Omnisound project group, “Omnisound: V1 prototype documentation,” Internal project README, desktop-first high-fidelity prototype.

[3] J. Åhlander, “Projekt – användargränssnitt: Människa-datorinteraktion,” Mid Sweden University, course project instructions, Oct. 2023, dated October 10, 2023; local file: P3 Projekt.pdf.

[4] Omnisound project group, “Omnisound – iteration 2: Prototype documentation,” Internal project README, section: Iteration 2 redesign.

[5] O. A. Aljabi, Y. Dawli, and S. Chavoshi Rad, “Alternative assignment for project session 2,” Unpublished data collection plan, Group 1, DT168G/IU127G, submitted two-page PDF; planned interview roles on p. 2.